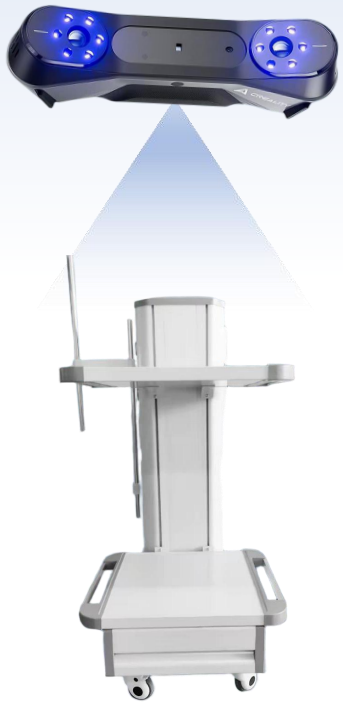
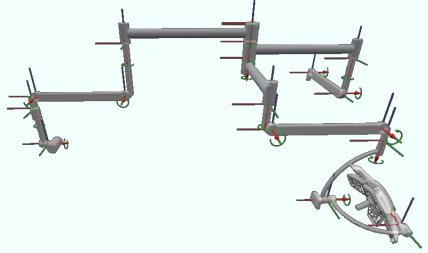
Proof-of-Concept Construction Pipeline

1 Physical asset scanning



2 Model definition and simulation scene setup

2a URDF definition with kinematics



2b Scene setup in Isaac Sim



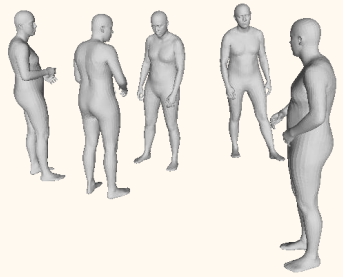
3 Human capture and mesh reconstruction

3a Dual-view capture

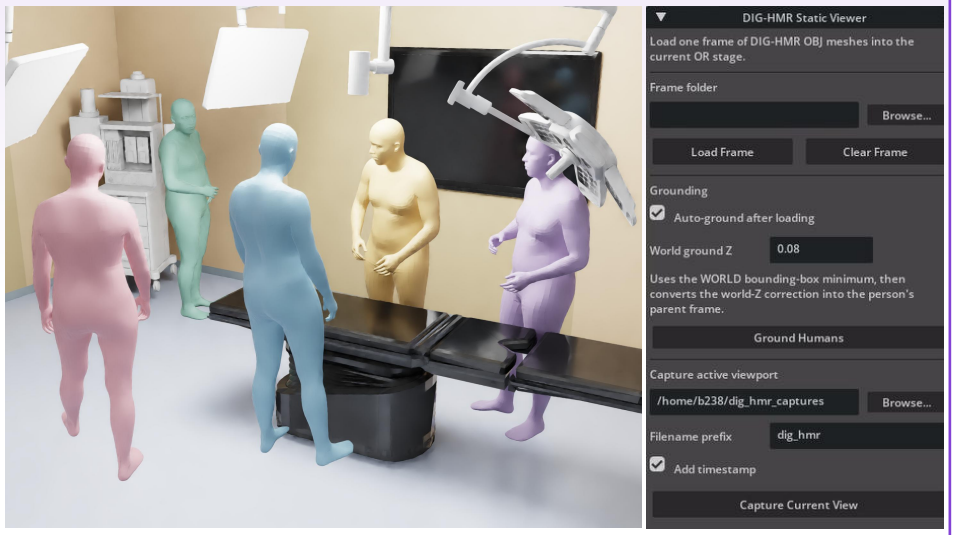




3b DIG-HMR



4 Scene-human integration and plugin-based proof-of-concept



- Relistic spatial layout
- Custom simulator plugin
  - Set Ground height and alignment
  - Load targert-frame human meshes